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Multilayer high precision PCB's with impedance control

JLCPCB can produce High-precision multilayer board with capabilities listed in below table.

Min. Trace width/Spacing Min. Via Min. BGA

3.5mil 0.2mm 0.25mm

It is very convenient for customers to conduct an impedance matching design, according to JLCPCB's laminated structure and related parameters. The PCB will be strictly produced in accordance with the following stackup.

Controlled Impedance PCB Parameters and Stackup

1.Prepreg dielectric constant:

Prepreg type Dielectric constant

7628	4.4
3313	4.1
1080	3.91
2116	4.16

2.Solder mask Parameters

Coating Above Substrate C1	Coating Above Trace C2	Coating Between Traces C3	Coating Dielectric CEr
1.2mil	0.6mil	1.2mil	3.8

3. Core dielectric constant

Core dielectric constant

4.6

For your convenience, we have designed an [Impedance Calculator](#) to help you calculate the impedance and the trace width you require.

4-Layer Impedance Control Stackup

Thickness

0.8mm 1.0mm 1.2mm 1.6mm 2.0mm

Outer Copper Weight

1oz 2oz

inner Copper Weight

0.5oz 1oz 2oz

- 1) No requirement Stackup

Layer

Material Type

Thickness

Top Layer

Copper

0.035mm

Prepreg

7628*1

0.21040mm

Inner Layer L2

Core

Inner Layer L3

Copper

Core

Copper

0.0152mm

1.065mm

0.0152mm

1.1mm H/HOZ with copper

Prepreg

7628*1

0.21040mm

Bottom Layer

Copper

0.035mm

- 2) JLC04161H-7628 Stackup

Layer

Material Type

Thickness

Top Layer

Copper

0.035mm

Prepreg

7628*1

0.21040mm

Inner Layer L2

Core
Inner Layer L3
Copper
Core
Copper
0.0152mm
1.065mm
0.0152mm
1.1mm H/HOZ with copper
Prepreg
7628*1
0.21040mm
Bottom Layer
Copper
0.035mm

- 3) JLC04161H-3313 Stackup

Layer
Material Type
Thickness
Top Layer
Copper
0.035mm
Prepreg
3313*1
0.09940mm
Inner Layer L2
Core
Inner Layer L3
Copper
Core
Copper
0.0152mm
1.265mm
0.0152mm
1.3mm H/HOZ with copper
Prepreg
3313*1
0.09940mm
Bottom Layer
Copper
0.035mm

- 4) JLC04161H-1080 Stackup

Layer
Material Type
Thickness
Top Layer
Copper
0.035mm
Prepreg

1080*1
0.07640mm
Inner Layer L2
Core
Inner Layer L3
Copper
Core
Copper
0.0152mm
1.265mm
0.0152mm
1.3mm H/HOZ with copper
Prepreg
1080*1
0.07640mm
Bottom Layer
Copper
0.035mm

- 5) JLC04161H-7628A Stackup

Layer
Material Type
Thickness
Top Layer
Copper
0.035mm
Prepreg
7628*1
0.218mm
Prepreg
1080*1
0.07640mm
Inner Layer L2
Core
Inner Layer L3
Copper
Core
Copper
0.0152mm
0.865mm
0.0152mm
0.9mm H/HOZ with copper
Prepreg
1080*1
0.07640mm
Prepreg
7628*1
0.218mm
Bottom Layer
Copper
0.035mm

- 6) JLC04161H-3313A Stackup

Layer
Material Type
Thickness
Top Layer
Copper
0.035mm
Prepreg
3313*1
0.107mm
Prepreg
3313*1
0.09940mm
Inner Layer L2
Core
Inner Layer L3
Copper
Core
Copper
0.0152mm
1.065mm
0.0152mm
1.1mm H/HOZ with copper
Prepreg
3313*1
0.09940mm
Prepreg
3313*1
0.107mm
Bottom Layer
Copper
0.035mm

- 7) JLC04161H-1080A Stackup

Layer
Material Type
Thickness
Top Layer
Copper
0.035mm
Prepreg
1080*1
0.084mm
Prepreg
1080*1
0.07640mm
Inner Layer L2
Core
Inner Layer L3
Copper
Core

Copper
0.0152mm
1.065mm
0.0152mm
1.1mm H/HOZ with copper
Prepreg
1080*1
0.07640mm
Prepreg
1080*1
0.084mm
Bottom Layer
Copper
0.035mm

- 8) JLC04161H-7628B Stackup

Layer
Material Type
Thickness
Top Layer
Copper
0.035mm
Prepreg
7628*1
0.218mm
Prepreg
7628*1
0.218mm
Prepreg
2116*1
0.11640mm
Inner Layer L2
Core
Inner Layer L3
Copper
Core
Copper
0.0152mm
0.4mm
0.0152mm
0.4mm H/HOZ without copper
Prepreg
2116*1
0.11640mm
Prepreg
7628*1
0.218mm
Prepreg
7628*1
0.218mm
Bottom Layer
Copper

0.035mm

- 9) JLC04161H-2116A Stackup

Layer

Material Type

Thickness

Top Layer

Copper

0.035mm

Prepreg

2116*1

0.124mm

Prepreg

7628*1

0.218mm

Prepreg

2116*1

0.11640mm

Inner Layer L2

Core

Inner Layer L3

Copper

Core

Copper

0.0152mm

0.6mm

0.0152mm

0.6mm H/HOZ without copper

Prepreg

2116*1

0.11640mm

Prepreg

7628*1

0.218mm

Prepreg

2116*1

0.124mm

Bottom Layer

Copper

0.035mm

- 10) JLC04161H-2116B Stackup

Layer

Material Type

Thickness

Top Layer

Copper

0.035mm

Prepreg

2116*1

0.124mm

Prepreg
7628*1
0.21040mm
Inner Layer L2
Core
Inner Layer L3
Copper
Core
Copper
0.0152mm
0.865mm
0.0152mm
0.9mm H/HOZ with copper
Prepreg
7628*1
0.21040mm
Prepreg
2116*1
0.124mm
Bottom Layer
Copper
0.035mm

- 11) JLC04161H-2116C Stackup

Layer
Material Type
Thickness
Top Layer
Copper
0.035mm
Prepreg
2116*1
0.124mm
Prepreg
1080*1
0.07640mm
Inner Layer L2
Core
Inner Layer L3
Copper
Core
Copper
0.0152mm
1.065mm
0.0152mm
1.1mm H/HOZ with copper
Prepreg
1080*1
0.07640mm
Prepreg
2116*1
0.124mm

Bottom Layer
Copper
0.035mm

- 12) JLC04161H-7628G Stackup

Layer
Material Type
Thickness
Top Layer
Copper
0.035mm
Prepreg
7628*1
0.218mm
Prepreg
7628*1
0.218mm
Prepreg
1080*1
0.07640mm
Inner Layer L2
Core
Inner Layer L3
Copper
Core
Copper
0.0152mm
0.5mm
0.0152mm
0.5mm H/HOZ without copper
Prepreg
1080*1
0.07640mm
Prepreg
7628*1
0.218mm
Prepreg
7628*1
0.218mm
Bottom Layer
Copper
0.035mm

- 13) JLC04161H-2116 Stackup

Layer
Material Type
Thickness
Top Layer
Copper
0.035mm
Prepreg

2116*1
0.11640mm
Inner Layer L2
Core
Inner Layer L3
Copper
Core
Copper
0.0152mm
1.265mm
0.0152mm
1.3mm H/HOZ with copper
Prepreg
2116*1
0.11640mm
Bottom Layer
Copper
0.035mm

- 14) JLC04161H-7628E Stackup

Layer
Material Type
Thickness
Top Layer
Copper
0.035mm
Prepreg
7628*1
0.218mm
Prepreg
7628*1
0.21040mm
Inner Layer L2
Core
Inner Layer L3
Copper
Core
Copper
0.0152mm
0.6mm
0.0152mm
0.6mm H/HOZ without copper
Prepreg
7628*1
0.21040mm
Prepreg
7628*1
0.218mm
Bottom Layer
Copper
0.035mm

- 15) JLC04161H-2116D Stackup

Layer
Material Type
Thickness
Top Layer
Copper
0.035mm
Prepreg
2116*1
0.124mm
Prepreg
7628*1
0.21040mm
Inner Layer L2
Core
Inner Layer L3
Copper
Core
Copper
0.0152mm
0.7mm
0.0152mm
0.7mm H/HOZ without copper
Prepreg
7628*1
0.21040mm
Prepreg
2116*1
0.124mm
Bottom Layer
Copper
0.035mm

- 16) JLC04161H-7628F Stackup

Layer
Material Type
Thickness
Top Layer
Copper
0.035mm
Prepreg
7628*1
0.218mm
Prepreg
7628*1
0.218mm
Prepreg
7628*1
0.21040mm
Inner Layer L2
Core

Inner Layer L3

Copper

Core

Copper

0.0152mm

0.25mm

0.0152mm

0.25mm H/HOZ without copper

Prepreg

7628*1

0.21040mm

Prepreg

7628*1

0.218mm

Prepreg

7628*1

0.218mm

Bottom Layer

Copper

0.035mm

- 17) JLC04161H-2116E Stackup

Layer

Material Type

Thickness

Top Layer

Copper

0.035mm

Prepreg

2116*1

0.124mm

Prepreg

2116*1

0.11640mm

Inner Layer L2

Core

Inner Layer L3

Copper

Core

Copper

0.0152mm

0.865mm

0.0152mm

0.9mm H/HOZ with copper

Prepreg

2116*1

0.11640mm

Prepreg

2116*1

0.124mm

Bottom Layer

Copper

0.035mm

- 18) JLC04161H-7628C Stackup

Layer

Material Type

Thickness

Top Layer

Copper

0.035mm

Prepreg

7628*1

0.218mm

Prepreg

7628*1

0.218mm

Prepreg

7628*1

0.21040mm

Inner Layer L2

Core

Inner Layer L3

Copper

Core

Copper

0.0152mm

0.15mm

0.0152mm

0.15mm H/HOZ without copper

Prepreg

7628*1

0.21040mm

Prepreg

7628*1

0.218mm

Prepreg

7628*1

0.218mm

Bottom Layer

Copper

0.035mm

6-Layer Impedance Control Stackup

Thickness

1.2mm 1.6mm 2.0mm

Outer Copper Weight

1oz 2oz

inner Copper Weight

0.5oz 1oz 2oz

- 1) No requirement Stackup

Layer
Material Type
Thickness
Top Layer
Copper
0.035mm
Prepreg
3313*1
0.09940mm
Inner Layer L2
Core
Inner Layer L3
Copper
Core
Copper
0.0152mm
0.55mm
0.0152mm
0.55mm H/Hwithout copper
Prepreg
2116*1
0.11640mm
Inner Layer L4
Core
Inner Layer L5
Copper
Core
Copper
0.0152mm
0.55mm
0.0152mm
0.55mm H/Hwithout copper
Prepreg
3313*1
0.09940mm
Bottom Layer
Copper
0.035mm

- 2) JLC06161H-3313 Stackup

Layer
Material Type
Thickness
Top Layer
Copper
0.035mm
Prepreg
3313*1

0.09940mm
Inner Layer L2
Core
Inner Layer L3
Copper
Core
Copper
0.0152mm
0.55mm
0.0152mm
0.55mm H/Hwithout copper
Prepreg
2116*1
0.1088mm
Inner Layer L4
Core
Inner Layer L5
Copper
Core
Copper
0.0152mm
0.55mm
0.0152mm
0.55mm H/Hwithout copper
Prepreg
3313*1
0.09940mm
Bottom Layer
Copper
0.035mm

- 3) JLC06161H-7628 Stackup

Layer
Material Type
Thickness
Top Layer
Copper
0.035mm
Prepreg
7628*1
0.21040mm
Inner Layer L2
Core
Inner Layer L3
Copper
Core
Copper
0.0152mm
0.4mm
0.0152mm
0.4mm H/HOZ without copper
Prepreg

7628*1
0.2028mm
Inner Layer L4
Core
Inner Layer L5
Copper
Core
Copper
0.0152mm
0.4mm
0.0152mm
0.4mm H/HOZ without copper
Prepreg
7628*1
0.21040mm
Bottom Layer
Copper
0.035mm

- 4) JLC06161H-7628D Stackup

Layer
Material Type
Thickness
Top Layer
Copper
0.035mm
Prepreg
7628*1
0.216mm
Prepreg
7628*1
0.20840mm
Inner Layer L2
Core
Inner Layer L3
Copper
Core
Copper
0.0152mm
0.2mm
0.0152mm
0.2mm H/HOZ without copper
Prepreg
7628*1
0.20080mm
Inner Layer L4
Core
Inner Layer L5
Copper
Core
Copper
0.0152mm

0.2mm
0.0152mm
0.2mm H/HOZ without copper
Prepreg
7628*1
0.20840mm
Prepreg
7628*1
0.216mm
Bottom Layer
Copper
0.035mm

- 5) JLC06161H-1080 Stackup

Layer
Material Type
Thickness
Top Layer
Copper
0.035mm
Prepreg
1080*1
0.07640mm
Inner Layer L2
Core
Inner Layer L3
Copper
Core
Copper
0.0152mm
0.55mm
0.0152mm
0.55mm H/Hwithout copper
Prepreg
7628*1
0.21040mm
Inner Layer L4
Core
Inner Layer L5
Copper
Core
Copper
0.0152mm
0.55mm
0.0152mm
0.55mm H/Hwithout copper
Prepreg
1080*1
0.07640mm
Bottom Layer
Copper
0.035mm

- 6) JLC06161H-2116A Stackup

Layer
Material Type
Thickness
Top Layer
Copper
0.035mm
Prepreg
2116*1
0.11640mm
Inner Layer L2
Core
Inner Layer L3
Copper
Core
Copper
0.0152mm
0.13mm
0.0152mm
0.13mm H/HOZ without copper
Prepreg
2116*1
0.11640mm
Core
Core
0.7mm
0.7 H/HOZ without copper
Prepreg
2116*1
0.11640mm
Inner Layer L4
Core
Inner Layer L5
Copper
Core
Copper
0.0152mm
0.13mm
0.0152mm
0.13mm H/HOZ without copper
Prepreg
2116*1
0.11640mm
Bottom Layer
Copper
0.035mm

- 7) JLC06161H-1080A Stackup

Layer
Material Type
Thickness

Top Layer
Copper
0.035mm
Prepreg
1080*1
0.07640mm
Inner Layer L2
Core
Inner Layer L3
Copper
Core
Copper
0.0152mm
0.6mm
0.0152mm
0.6mm H/HOZ without copper
Prepreg
3313*1
0.09940mm
Inner Layer L4
Core
Inner Layer L5
Copper
Core
Copper
0.0152mm
0.6mm
0.0152mm
0.6mm H/HOZ without copper
Prepreg
1080*1
0.07640mm
Bottom Layer
Copper
0.035mm

- 8) JLC06161H-3313C Stackup

Layer
Material Type
Thickness
Top Layer
Copper
0.035mm
Prepreg
3313*1
0.107mm
Prepreg
1080*1
0.07640mm
Inner Layer L2
Core
Inner Layer L3

Copper
Core
Copper
0.0152mm
0.4mm
0.0152mm
0.4mm H/HOZ without copper
Prepreg
1080*1
0.07640mm
Prepreg
1080*1
0.07640mm
Inner Layer L4
Core
Inner Layer L5
Copper
Core
Copper
0.0152mm
0.4mm
0.0152mm
0.4mm H/HOZ without copper
Prepreg
1080*1
0.07840mm
Prepreg
3313*1
0.107mm
Bottom Layer
Copper
0.035mm

- 9) JLC06161H-1080B Stackup

Layer
Material Type
Thickness
Top Layer
Copper
0.035mm
Prepreg
1080*1
0.07640mm
Inner Layer L2
Core
Inner Layer L3
Copper
Core
Copper
0.0152mm
0.1mm
0.0152mm

0.10mm H/HOZ without copper

Prepreg

7628*1

0.21040mm

Core

Core

0.7mm

0.7 H/HOZ without copper

Prepreg

7628*1

0.21040mm

Inner Layer L4

Core

Inner Layer L5

Copper

Core

Copper

0.0152mm

0.1mm

0.0152mm

0.10mm H/HOZ without copper

Prepreg

1080*1

0.07640mm

Bottom Layer

Copper

0.035mm

- 10) JLC06161H-3313E Stackup

Layer

Material Type

Thickness

Top Layer

Copper

0.035mm

Prepreg

3313*1

0.09940mm

Inner Layer L2

Core

Inner Layer L3

Copper

Core

Copper

0.0152mm

0.1mm

0.0152mm

0.10mm H/HOZ without copper

Prepreg

7628*1

0.21040mm

Core

Core
0.7mm
0.7 H/HOZ without copper
Prepreg
7628*1
0.21040mm
Inner Layer L4
Core
Inner Layer L5
Copper
Core
Copper
0.0152mm
0.1mm
0.0152mm
0.10mm H/HOZ without copper
Prepreg
3313*1
0.09940mm
Bottom Layer
Copper
0.035mm

- 11) JLC06161H-2116B Stackup

Layer
Material Type
Thickness
Top Layer
Copper
0.035mm
Prepreg
2116*1
0.11640mm
Inner Layer L2
Core
Inner Layer L3
Copper
Core
Copper
0.0152mm
0.5mm
0.0152mm
0.5mm H/HOZ without copper
Prepreg
1080*1
0.07640mm
Prepreg
1080*1
0.07640mm
Inner Layer L4
Core
Inner Layer L5

Copper
Core
Copper
0.0152mm
0.5mm
0.0152mm
0.5mm H/HOZ without copper
Prepreg
2116*1
0.11640mm
Bottom Layer
Copper
0.035mm

- 12) JLC06161H-2116 Stackup

Layer
Material Type
Thickness
Top Layer
Copper
0.035mm
Prepreg
2116*1
0.127mm
Prepreg
2313*1
0.09640mm
Inner Layer L2
Core
Inner Layer L3
Copper
Core
Copper
0.0152mm
0.3mm
0.0152mm
0.30mm H/HOZ without copper
Prepreg
7628*1
0.20840mm
Prepreg
7628*1
0.20840mm
Inner Layer L4
Core
Inner Layer L5
Copper
Core
Copper
0.0152mm
0.3mm
0.0152mm

0.30mm H/HOZ without copper
Prepreg
2313*1
0.09640mm
Prepreg
2116*1
0.127mm
Bottom Layer
Copper
0.035mm

- 13) JLC06161H-7628B Stackup

Layer
Material Type
Thickness
Top Layer
Copper
0.035mm
Prepreg
7628*1
0.218mm
Prepreg
1080*1
0.07640mm
Inner Layer L2
Core
Inner Layer L3
Copper
Core
Copper
0.0152mm
0.35mm
0.0152mm
0.35mm H/HOZ without copper
Prepreg
1080*1
0.07640mm
Prepreg
1080*1
0.07640mm
Inner Layer L4
Core
Inner Layer L5
Copper
Core
Copper
0.0152mm
0.35mm
0.0152mm
0.35mm H/HOZ without copper
Prepreg
1080*1

0.07640mm
Prepreg
7628*1
0.218mm
Bottom Layer
Copper
0.035mm

- 14) JLC06161H-3313D Stackup

Layer
Material Type
Thickness
Top Layer
Copper
0.035mm
Prepreg
3313*1
0.107mm
Prepreg
1080*1
0.07640mm
Inner Layer L2
Core
Inner Layer L3
Copper
Core
Copper
0.0152mm
0.25mm
0.0152mm
0.25mm H/HOZ without copper
Prepreg
7628*1
0.21040mm
Prepreg
2116*1
0.124mm
Prepreg
7628*1
0.21040mm
Inner Layer L4
Core
Inner Layer L5
Copper
Core
Copper
0.0152mm
0.25mm
0.0152mm
0.25mm H/HOZ without copper
Prepreg
1080*1

0.07640mm
Prepreg
3313*1
0.107mm
Bottom Layer
Copper
0.035mm

- 15) JLC06161H-2116C Stackup

Layer
Material Type
Thickness
Top Layer
Copper
0.035mm
Prepreg
2116*1
0.127mm
Prepreg
2116*1
0.11940mm
Inner Layer L2
Core
Inner Layer L3
Copper
Core
Copper
0.0152mm
0.3mm
0.0152mm
0.30mm H/HOZ without copper
Prepreg
2116*1
0.11940mm
Prepreg
2116*1
0.127mm
Prepreg
2116*1
0.11940mm
Inner Layer L4
Core
Inner Layer L5
Copper
Core
Copper
0.0152mm
0.3mm
0.0152mm
0.30mm H/HOZ without copper
Prepreg
2116*1

0.11940mm
Prepreg
2116*1
0.127mm
Bottom Layer
Copper
0.035mm

- 16) JLC06161H-7628A Stackup

Layer
Material Type
Thickness
Top Layer
Copper
0.035mm
Prepreg
7628*1
0.218mm
Prepreg
7628*1
0.218mm
Prepreg
1080*1
0.07640mm
Inner Layer L2
Core
Inner Layer L3
Copper
Core
Copper
0.0152mm
0.1mm
0.0152mm
0.10mm H/HOZ without copper
Prepreg
2116*1
0.11640mm
Prepreg
2116*1
0.11640mm
Inner Layer L4
Core
Inner Layer L5
Copper
Core
Copper
0.0152mm
0.1mm
0.0152mm
0.10mm H/HOZ without copper
Prepreg
1080*1

0.07640mm
Prepreg
7628*1
0.218mm
Prepreg
7628*1
0.218mm
Bottom Layer
Copper
0.035mm

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